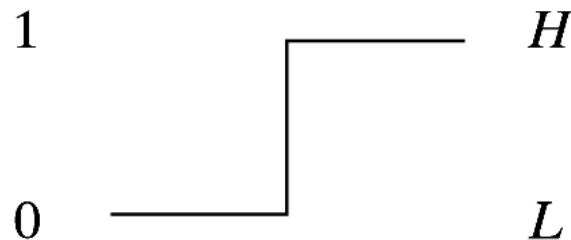


Digital systems Overview

- Digital systems is one where both the input and the output are digital signals
- Digital Circuits are widely used since they
 - Operate in one of the two states ON/OFF, resulting in simple circuit operation.
 - Can be realized as **integrated circuits** which are highly reliable, extremely small in size, and cost very less.

Digital Circuits

- Involves circuit that has two possible states
i.e., Binary System
- The operation of the circuit can be described in terms of its Voltage levels
 - There are only two voltage levels High or Low

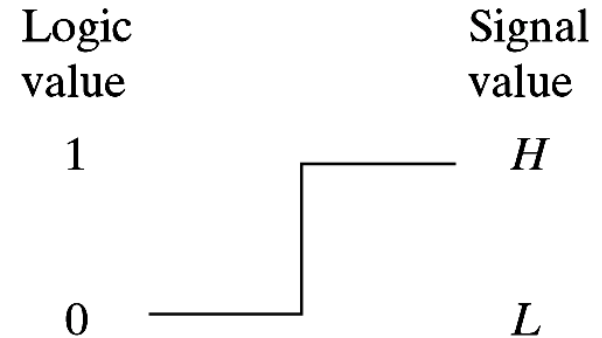


Digital Circuits

Can be of two types

- Positive logic

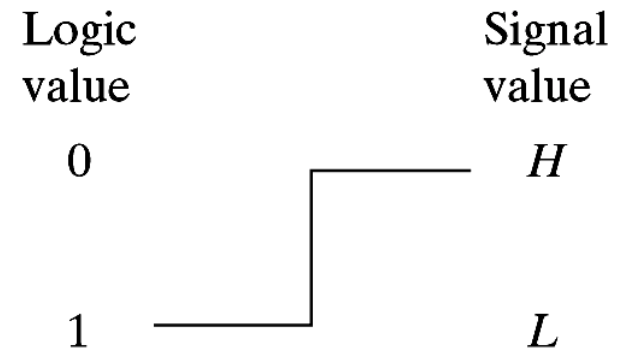
- Choose the high-level H as logic 1
- Choose the low-level L as logic 0



(a) Positive logic

- Negative logic

- Choose the high-level H as logic 0
- Choose the low-level L as logic 1



(b) Negative logic

Logic Gates

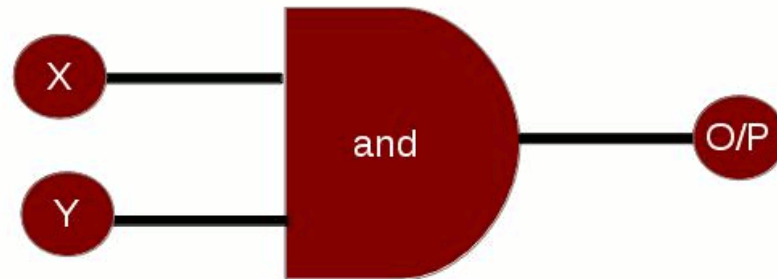
- Logic gates are electronic circuits elements that perform the corresponding logical operations
- *Properties of Logic Gates*
 - Logic gates handle binary signals and also generate binary signals
 - Logic gates can have more than two inputs

Basic Gates: AND Gate

- The AND gate will produce a 1 at the output O/P if both the inputs X and Y change to 1

and Gate Animation

X	Y	O/P
0	0	0
0	1	0
1	0	0
1	1	1

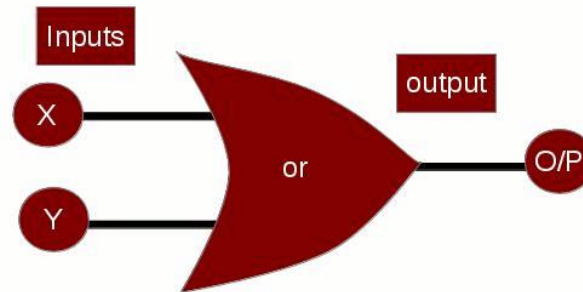


Basic Gates: OR Gate

- The OR gate will output 1 at Q if either of the logic states of inputs A and B change to 1

or Gate Animation

X	Y	O/P
0	0	0
0	1	1
1	0	1
1	1	1

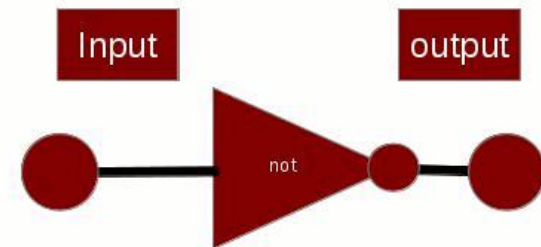


Basic Gates: NOT Gate

- The NOT gate reverses the input.
- It is also known as INVERTER

not Gate Animation

X	O/P
0	1
1	0

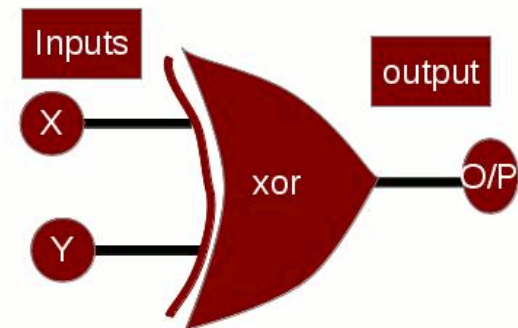


Ex-OR Gate

- The output of an Ex-OR gate goes “HIGH” only when both of its input terminals are at “DIFFERENT” logic levels with respect to each other
- It implements the function ***“A OR B but NOT both”***

xor Gate Animation

X	Y	O/P
0	0	0
0	1	1
1	0	1
1	1	0



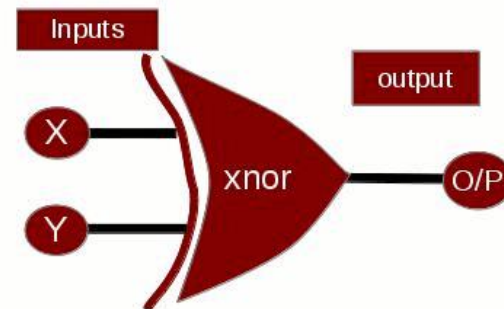
ODD number of 1's at the input results in output high

Ex-NOR Gate

- The output of an Ex-NOR gate goes “HIGH” when both of its input terminals are at “SAME” logic levels with respect to each other
- It is also known as equivalence gate

xnor Gate Animation

X	Y	O/P
0	0	1
0	1	0
1	0	0
1	1	1



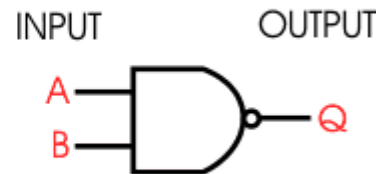
EVEN number of 1's at the input results in output high

NAND Gate

- The NAND gate has the opposite outputs to the AND gate. It is NOT-AND.

NAND gate

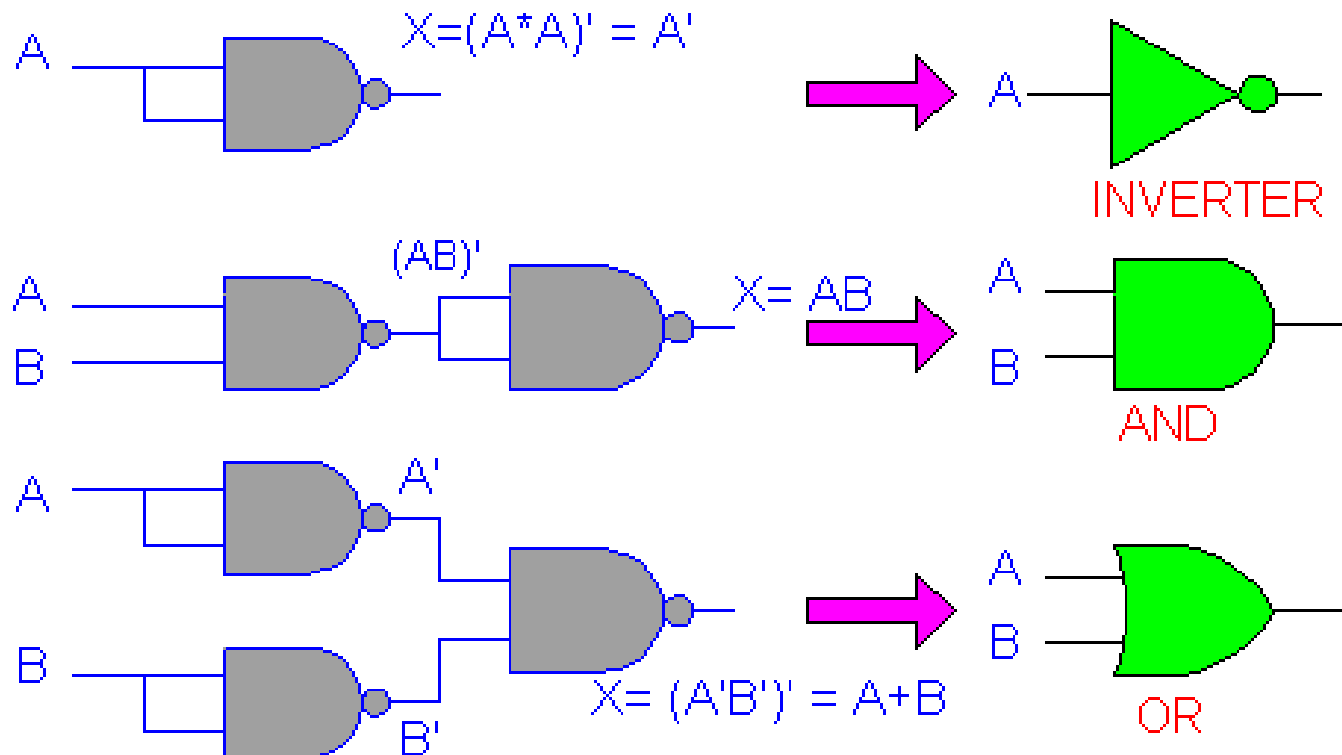
A	B	Q
0	0	1



NAND & NOR are Universal Gates

NAND and NOR gates are more popular than AND and OR gates as they are more easily constructed with transistor circuits

Universality of NAND Gate

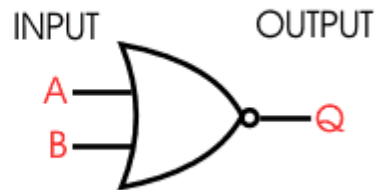


NOR Gate

- The NOR gate has the opposite outputs to the OR gate. It is NOT-OR.

NOR gate

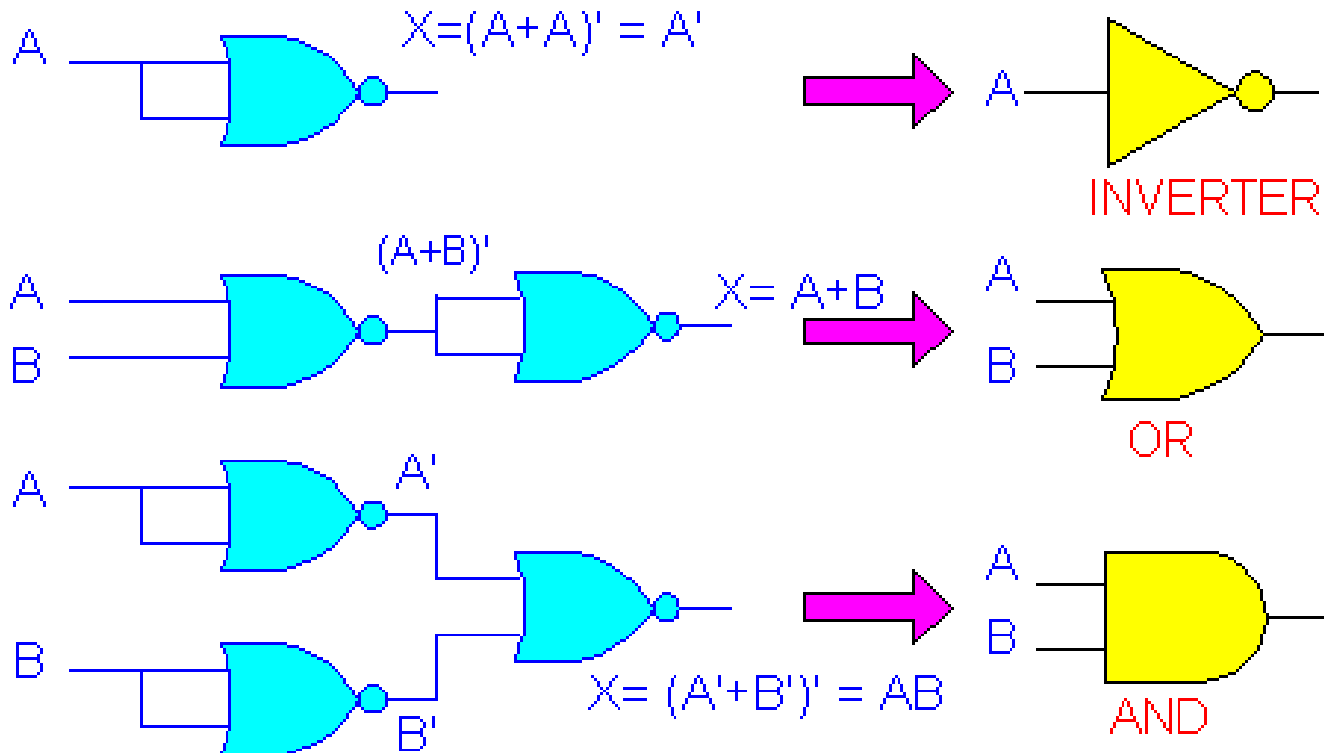
A	B	Q
0	0	1



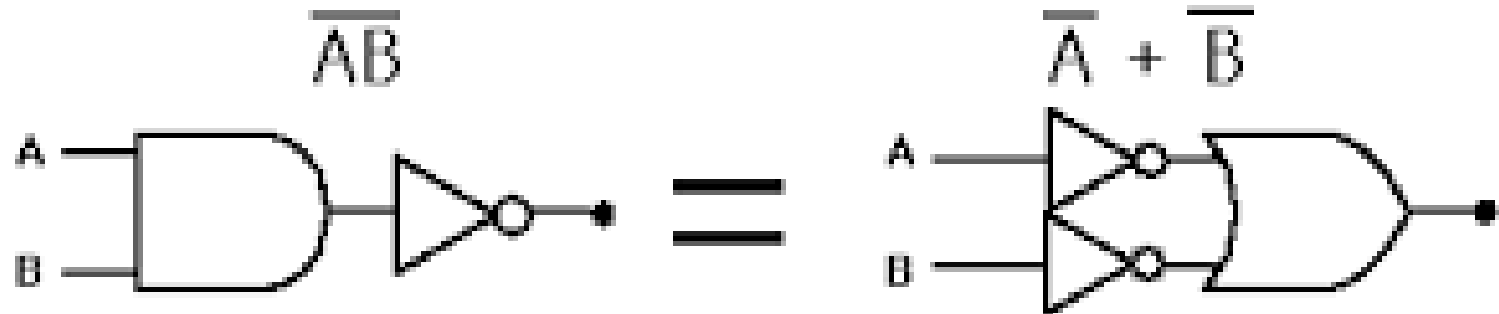
NAND & NOR are Universal Gates

NAND and NOR gates are more popular than AND and OR gates as they are more easily constructed with transistor circuits

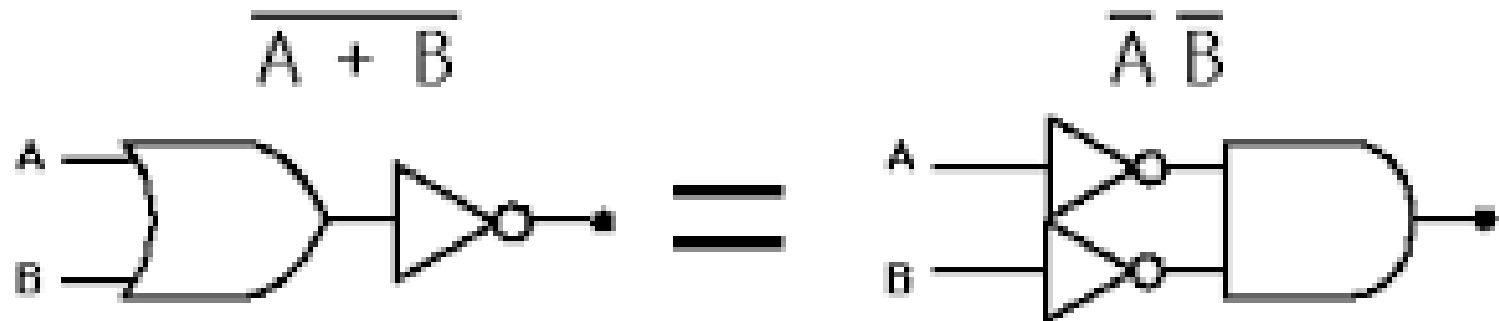
Universality of NOR Gate



DeMorgan's Theorem



A NAND gate is equivalent to an inversion followed by an OR



A NOR gate is equivalent to an inversion followed by an AND

Boolean Laws

Law/Theorem	Law of Addition	Law of Multiplication
Identity Law	$x + 0 = x$	$x \cdot 1 = x$
Complement Law	$x + x' = 1$	$x \cdot x' = 0$
Idempotent Law	$x + x = x$	$x \cdot x = x$
Dominant Law	$x + 1 = 1$	$x \cdot 0 = 0$
Involution Law	$(x')' = x$	
Commutative Law	$x + y = y + x$	$x \cdot y = y \cdot x$
Associative Law	$x + (y + z) = (x + y) + z$	$x \cdot (y \cdot z) = (x \cdot y) \cdot z$
Distributive Law	$x \cdot (y + z) = x \cdot y + x \cdot z$	$x + y \cdot z = (x + y) \cdot (x + z)$
Demorgan's Law	$(x + y)' = x' \cdot y'$	$(x \cdot y)' = x' + y'$
Absorption Law	$x + (x \cdot y) = x$	$x \cdot (x + y) = x$

Thank You